Object Oriented Programming Lab (CSEG1044) Experiment-01&02: Java Programming Environment + Input Handling (Student Handout — Before Submission)

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1 About this handout

This is a **pre-submission** student handout. It provides **requirements** and **hints**, but not full solutions. Teacher model solutions are kept privately.

2 Experiment 01 (Syllabus)

- **Objective:** To understand the Java programming environment, program structure, compilation, and execution process.
- **Problem Statement:** Design and implement a basic Java program demonstrating class declaration, `main()` method, and standard output statements.
- **Expected Outcome:** Ability to write, compile, and execute a basic Java program.

3 Experiment 02 (Syllabus)

- **Objective:** To understand different methods of input handling in Java.

- **Problem Statement:** Develop a Java program that accepts input using standard input methods and command line arguments.
- **Expected Outcome:** Ability to handle input from multiple sources.

4 What you must do (minimum requirements)

4.1 Task A — Java setup + first program (Exp-01)

1. Verify installation (attach output screenshots):
   - `java -version`
   - `javac -version`
2. Create `HelloUPES.java` that prints:
   - a welcome message,
   - Java version using `System.getProperty("java.version")`,
   - number of command line arguments.
3. Compile and run:
   - `javac HelloUPES.java`
   - `java HelloUPES`
   - `java HelloUPES A B`

4.2 Task B — Keyboard input + command line args (Exp-02)

1. Create `InputArgsDemo.java` that:
   - reads `bonusPercent` from command line arguments,
   - reads `name` and `baseSalary` using `Scanner`,
   - computes `finalSalary = baseSalary + baseSalary * bonusPercent / 100`,
   - prints the final salary using formatted output.
2. Validations (mandatory):
   - check `args.length` before accessing `args[0]`,
   - handle invalid bonus argument (`NumberFormatException`),
   - handle invalid salary input (`InputMismatchException`).

5 Hints (read only if stuck)

- Command line arguments are always strings; parse numbers using `Integer.parseInt` or `Double.parseDouble`.
- Close `Scanner` using `finally` or try-with-resources.
- If you mix `nextInt()/nextDouble()` with `nextLine()`, remember the newline issue.

6 What to submit (MANDATORY)

Submit a single zipped folder containing:

- source code: `HelloUPES.java`, `InputArgsDemo.java`
- output screenshots or copied output
- observations (5–10 lines): compile-time vs runtime; what happens for invalid inputs

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed March 3, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed March 3, 2026).